

IRT-ZL Series, RS485 And Analog Signal Synchronous Output, Built-in line laser for rapid alignment, LCD Display

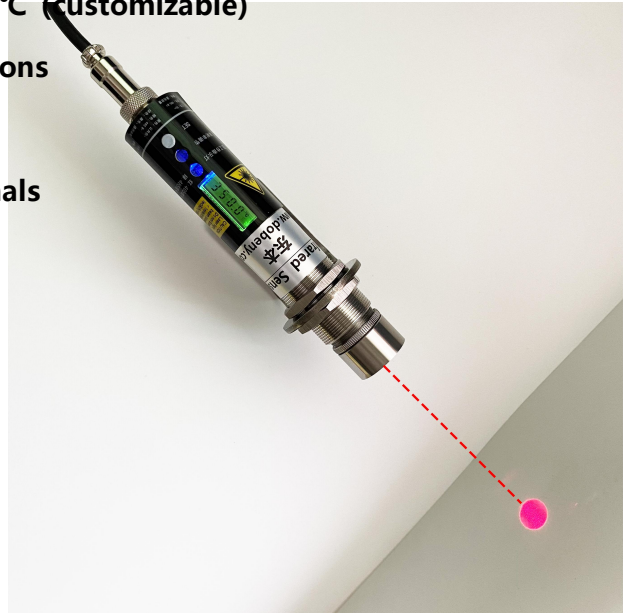
- **Temperature measurement range: 150°C~2000°C (customizable)**
- **Suitable for measuring metals and metal solutions**
- **Adjustable focal length (300mm-1000mm)**
- **Simultaneous output of RS485 and analog signals**
- **LCD temperature display**
- **Adjustable emissivity**

DOBENY Technology IRT/ZL.01 series Infrared laser Temperature Sensor has the function of built-in line laser for rapid alignment;

The LCD displays the current status, which can easily align the measurement target and indicate the best measurement distance;

The focus point can measure 3mm objects;

Various types of options (purgers, cooling protective jackets, etc.) are available to meet the requirements of various working conditions.



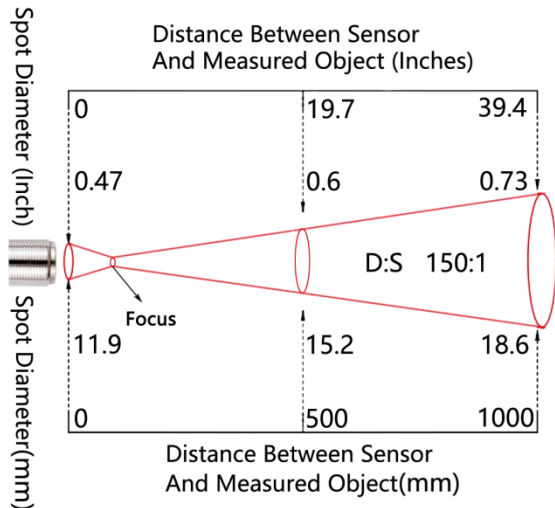
Details

Basic Details			
Temperature measurement range	150°C~1800°C	Response time	50ms (95% Response)
Output signal	RS485\4-20mA\0-5V	Spectral range	1.0μm、1.6μm、2.3μm
Accuracy	±1.2%	Supply	24 V DC
Repeatability	±1%	Field of View	120:1
Emissivity	0.100-1.099, step size 0.001	Maximum current/resistance	80mA/500Ω
Focal length	300-1000mm	Spot size	3mm

Appearance properties			
Material	Stainless Steel	Dimensions	Φ25×174mm
Weight	About 300g	Cable length	3m (Can be customized length)

Use environment			
Environmental (IP) rating	IP65	Ambient temperature	0°C~60°C
Humidity	10%-95%(No condensation)	Storage environment	-20°C~80°C

Field of View: 150:1

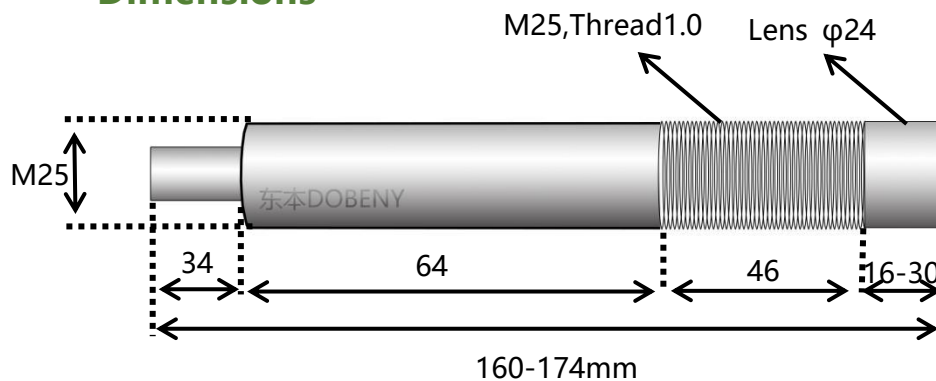


Note: The focus point distance and spot size are subject to the actual product.

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Dimensions



Wiring

Cable color	Function	Cable color	Function
Red	Current output 9-24V DC (+)	Black	Public negative -
	485 output 6-24V DC (+)	Green	The laser lights up the line, triggers the light, and it is not recommended to be always on
Yellow	Parameter setting line	Transparent	Shield
White	4-20mA or 0-20mA (output signal)	Blue	485 A+
Orange	485 B-		

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Focus on infrared temperature measurement

Other Instructions

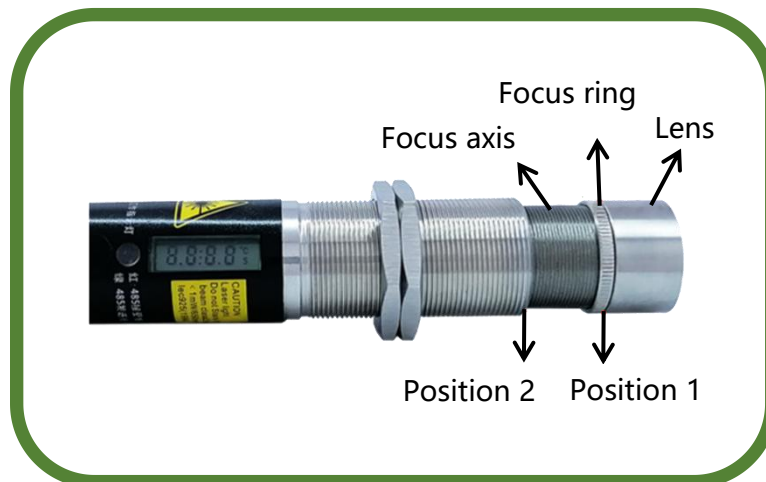
Focus method

The front section of the probe can be manually adjusted and the wheelbase is 10mm;

After the probe is fixedly installed, rotate the focus ring to position 1;

Manually rotate the probe lens until the laser spot is the brightest and smallest, which is the best measurement focal length;

Then rotate the focus ring to position 2 to lock the focus axis.



Determine the emissivity of an object

The intensity of infrared radiation emitted by an object depends on the temperature of the object and the radiation properties of the material on the surface of the object. We use the parameter emissivity (ϵ -Epsilon) to describe the ability of an object to radiate energy outward. The emissivity can range from 0 to 100%. What we usually refer to as a "black body" refers to an ideal source of radiation with an emissivity of 1.0, while a mirror typically has an emissivity of 0.1. If you select an emissivity that is too high when measuring temperature with an infrared thermometer, the temperature displayed by the thermometer will be lower than the true temperature of the object being measured - assuming that the temperature of the object being measured is higher than the ambient temperature.

Measurement errors caused by objects with low emissivity (reflective surfaces) due to interference from other external radiation or background objects (flames, heating systems, refractories), in this case to reduce measurement errors, install and protect the probe very carefully. Avoid reflected radiation sources.

To determine the unknown emissivity of an object, follow these steps

Use other contact meters to determine the true temperature of the object to be measured, then measure the object with an infrared thermometer, and adjust the emissivity settings so that the displayed value of the infrared thermometer is consistent with the actual temperature of the object.

For low temperature thermometers, a part of the surface of the object to be measured can be painted with black pigment with an emissivity of 0.98, the emissivity of the infrared thermometer can be adjusted to 0.98, the temperature of this part can be measured, and then the adjacent edges can be measured. temperature and adjust the emissivity until the displayed value matches the temperature just measured.

Typical emissivity

When none of the methods mentioned above can help you determine the emissivity of an object, you can use the emissivity table given in the attached table. This is only an average value, the actual material emissivity depends on the following factors: material temperature, measurement angle, surface geometry, material thickness, material surface structure (polished, oxidized, roughened, sandblasted), measured spectral range, transmission rate (such as thin films).